VEDAVYAS ROKALABOINA

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https://vedavyasrokalaboina.github.io/

PERSONAL PROFILE

Aeronautical Engineering MSc graduate with strong expertise in simulation, mathematical modelling, and computational analysis, developed through aerospace design and UAV development projects. Experienced in CFD, FEA, and performance modelling using ANSYS, MATLAB, and Python, with a proven ability to translate real-world systems into high-fidelity simulation environments. Skilled in analysing complex datasets, validating models, and optimising system performance through numerical methods. Passionate about applying simulation-driven engineering to high-performance environments such as motorsport, with a strong interest in vehicle dynamics, model correlation, and performance optimisation.

SKILLS

Simulation & Analysis: CFD (ANSYS Fluent), FEA, Numerical Modelling, Model Validation

Programming: MATLAB, Python

CAD & Tools: CATIA V5, SolidWorks, XFLR5, Open FOAM

Engineering Areas: Aerodynamics, Flight Mechanics, Performance Analysis

Soft Skills: Problem-solving, Team collaboration, Technical communication

EDUCATION

UNIVERSITY OF SOUTH WALES – Master`s degree | SEP 2024 – SEP 2025

MSc Aeronautical Engineering.

- Focused on aerodynamics, simulation, propulsion, and computational engineering
- Applied MATLAB and CFD tools to analyse and optimise engineering systems

INSTITUTE OF AERONAUTICAL ENGINEERING - Bachelor`s degree | DEC 2020 – APR 2024

Bachelor of Technology Aeronautical Engineering

- Strong grounding in aircraft design, structures, and aerodynamics
- Extensive use of simulation tools for performance evaluation and design validation

WORK EXPERIENCE

Technical Lead – UAV Development

Abhuday Bharat Projects Pvt Ltd | Mar 2024 – Sep 2025

- Designed UAV components using SolidWorks and CATIA
- Performed CFD and FEA analysis using ANSYS Fluent to optimise aerodynamic performance

- Developed performance models and analysed simulation data using MATLAB
- Supported prototype development and testing, improving system reliability
- Collaborated with multidisciplinary teams to deliver engineering solutions

Technical Lead – Aeronautical Projects (Student Team)

ARDC IARE | Apr 2022 – Feb 2024

- Led aeronautical projects involving CFD, structural analysis, and system modelling
- Developed simulation models to analyse aerodynamic and flight performance
- Guided team members in using ANSYS, MATLAB, and CAD tools
- Delivered technical reports and presentations based on analysis results

PROJECTS AND ACCOMPLISHMENTS

SAE INTERNATIONAL AND SAE NATIONAL AERO DESIGN CHALLENGE **JAN 2022 – MAR 2023**

- Led CFD, FEA, and CAD activities as Technical Lead for SAE Aero Design projects
- Achieved 15th place (International – USA 2023) and 3rd place (National – India 2022)

THRUST VECTOR CONTROL BY JET VANES **JUN 2023 – APR 2024**

- Conducted CFD analysis on multiple nozzle geometries
- Analysed high-speed flow characteristics and performance
- Identified optimal nozzle design based on simulation results

BUSINESS JET DESIGN, ANALYSIS AND SIMULATION **SEP 2024 – APR 2025**

- Designed an 8-passenger aircraft based on mission requirements
- Estimated take-off weight and key design parameters
- Used MATLAB for performance calculations and modelling
- Created CAD model and validated performance through simulation

INVESTIGATION OF THE AERODYNAMICS OF A BIO – INSPIRED BWB **APR 2025 – SEP 2025**

- Developed high-fidelity CFD models using ANSYS Fluent
- Conducted mesh convergence and validation studies
- Performed parametric analysis to optimise lift-to-drag ratio
- Improved aerodynamic efficiency through simulation insights

ACADEMIC REFERENCE

DR ROBERT WARREN – Lecturer in Aeronautical Engineering (USW Treforest campus)

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